

# NITKIT

## 3D Printing Defect Detection

Automated Visual Inspection using YOLO26 + Local LLM Troubleshooting

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Computer Vision

Deep Learning

FDM 3D Printing

Flask API

Ollama LLM



## THE PROBLEM

# Why Automate 3D Print Inspection?

### The Challenge

- FDM 3D printing is prone to defects that compromise quality.
- Manual inspection is slow, inconsistent, and hard to scale.

### The Solution

- Train YOLO26 to detect defects in real time.
- Pair it with an LLM to give instant fix suggestions.

## Project Objectives



Auto-detect defects



Classify defect types



LLM troubleshooting



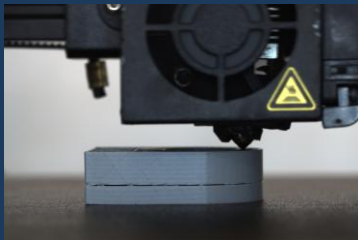
Future real-time pipeline

## 3 Types of Defects Detected

### Crack

AP = 0.940

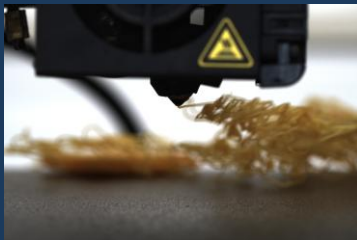
Visible fractures or breaks in the printed part. Usually caused by rapid cooling or layer adhesion failure. Highly distinctive — easy to detect.



### Spaghetti

AP = 0.500

Catastrophic failure where filament extrudes freely in tangled strands. Print has lost contact with the build plate. Very variable in appearance.



### Stringing

AP = 0.793

Thin filament threads between features during travel moves. Caused by too-high temperature or insufficient retraction. Fine, diffuse patterns.



# How the Model Was Built

2,058

Source Images  
(Roboflow)

×3

Augmented  
per image

640px

Training  
resolution

500

Training  
epochs

## Dataset Pipeline

### Roboflow Universe

3d-print-fail-detection (CC BY 4.0)

**Pre-processing:** Auto-orient · Resize to 512px ·  
Grayscale · Histogram equalisation

**Augmentation:** Random crop · Brightness  $\pm 15\%$  ·  
Exposure  $\pm 10\%$

## Training Config

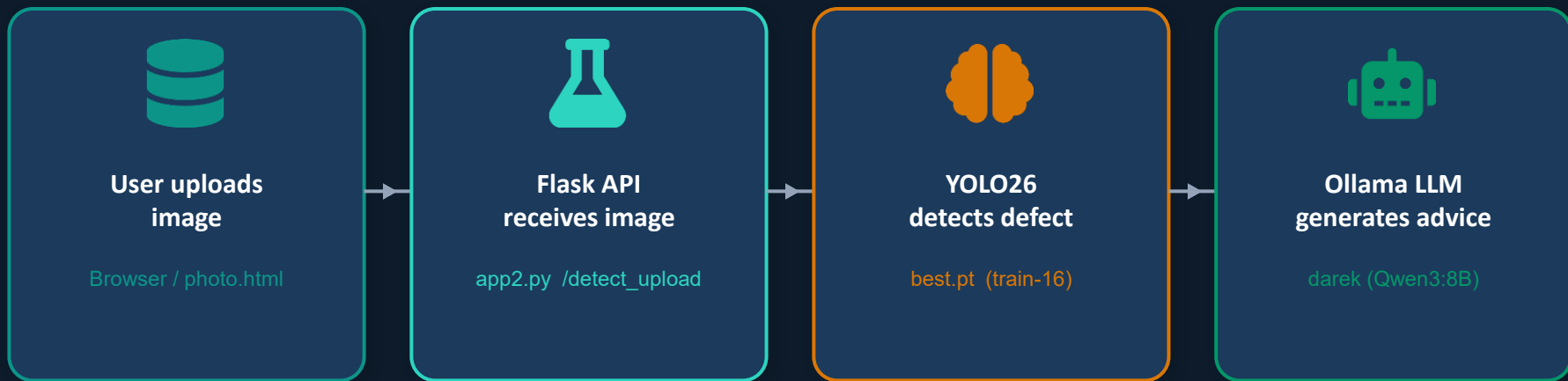
**Model:** yolo26n.pt (YOLO26 nano)

**Optimizer:** Auto · LR 0.01 · Momentum 0.937

**Device:** CUDA GPU (Laptop GPU RTX3060, 16GB RAM)

**Batch:** Auto · Image size: 640×640

# How It All Connects



Result returned to browser:



Annotated image  
with bounding boxes



Detected class  
label



LLM fix advice  
(≤5 sentences)

## RESULTS

# Model Performance

**74.5%**

Overall mAP@0.5

**0.75**

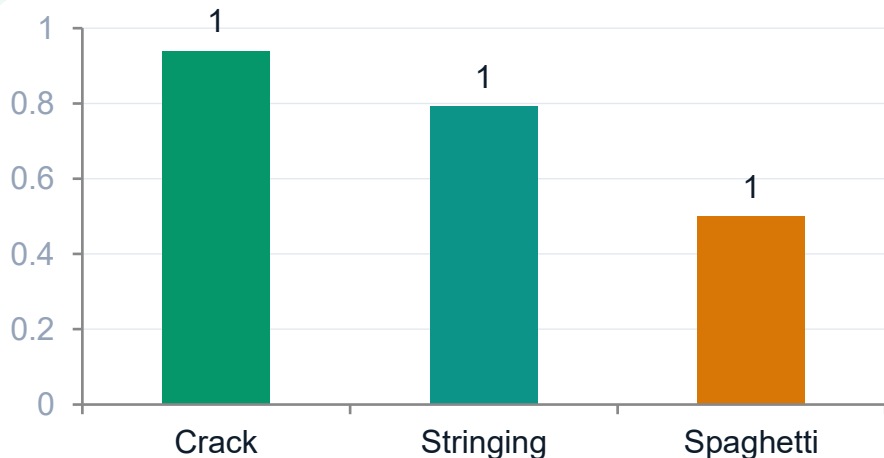
Best F1 Score

**0.628**

Optimal Threshold

**500**

Training Epochs



### Detection Rates

|           |            |     |
|-----------|------------|-----|
| Crack     | ✓ 141 X 9  | 94% |
| Stringing | ✓ 138 X 39 | 78% |
| Spaghetti | ✓ 15 X 13  | 54% |

*Key finding: most errors are 'missed as background', not confused with other defect classes.*

## Additive Manufacturing Detection

OBS Virtual Camera

Start Camera

Capture Photo

Feature 1: Defect capture  
機能1: 欠陥画像の撮影

Talk to Darek

Type a message for Darek...

Send

# Strengths & Limitations



### Strengths

- Excellent crack detection (AP 0.940)
- Good stringing accuracy (AP 0.793)
- LLM gives instant fix advice per defect type
- Fully local — no external API needed
- Lightweight YOLO26n — fast inference
- End-to-end web app, ready to deploy



### Limitations

- Spaghetti AP only 0.500 — needs more data
- Trained on one environment / lighting setup
- LLM prompt is fixed — not image-aware
- Object counter uses simple heuristic only
- requirements.txt has encoding issue (UTF-16)



# How can we improve from this?

### More Data

Collect more spaghetti samples. Add diverse printers, materials & lighting.

### Bigger Model

Try YOLO26s or YOLO26m variants for higher accuracy on complex classes.

### Smarter LLM

Pass bounding box coords & confidence to the LLM for specific advice.

### Live Dashboard

Integrate with OctoPrint / Klipper for real-time auto-cancel on failure.

**Conclusion** The project demonstrates that a YOLO26 + local LLM pipeline can effectively automate quality inspection for FDM 3D printing, reaching 74.5% mAP@0.5 with excellent crack detection. It provides a strong foundation for a production-ready, real-time inspection and troubleshooting system.

# Thank you for your attention

Any questions?